

DevopsEssentials(01CE0717)

Lab Manual 26-27

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Class: 7-EC1

Batch: A

Lab	Program	Signature	Marks
1	Build a branching model to help your team understand the Git workflow for faster integration of work.		
2	Use Jenkins to set up a distributed pipeline that will compile and test a Maven project on two different slave nodes respectively.		
3	Create Jenkins job and pipeline it from GitHub repo to run using build.		
4	Building a CI/CD Pipeline with Jenkins.		
5	Dockerizing Jenkins Pipeline.		
6	Deploy web app in AWS EC2 using Ansible.		
7	Deploy Angular/React/Java/Python or any other Application in Docker Container		
8	Use Jenkins to set up a distributed pipeline that will compile and test a Maven project on two different slave nodes respectively.		

Practical 1: Build a branching model to help your team understand the Git workflow for faster integration of work.

Step 1: Initialize Git Repository

```
mkdir git-branching-demo  
cd git-branching-demo  
git init
```

```
D:\>mkdir git-branching-demo  
  
D:\>cd git-branching-demo  
  
D:\git-branching-demo>git init  
Initialized empty Git repository in D:/git-branching-demo/.git/
```

Step 2:

Initial Commit

```
echo "# Git Branching Demo" > README.md  
git add .  
git commit -m "Initial commit"
```

```
D:\git-branching-demo>  
D:\git-branching-demo>echo "# Git Branching Demo" > README.md  
  
D:\git-branching-demo>git add .  
  
D:\git-branching-demo>git commit -m "Initial commit"  
[main (root-commit) cb00c0f] Initial commit  
1 file changed, 1 insertion(+)  
create mode 100644 README.md  
  
D:\git-branching-demo>
```

Step 3: Create Develop Branch

```
git branch develop  
git checkout develop  
git branch
```

```
D:\git-branching-demo>git branch develop  
  
D:\git-branching-demo>git checkout  
  
D:\git-branching-demo>git branch  
develop  
* main  
  
D:\git-branching-demo>
```

Step 4: Create Feature Branch

git checkout -b feature/login

```
D:\git-branching-demo>git checkout -b feature/login
Switched to a new branch 'feature/login'

D:\git-branching-demo>
```

Step 5: Work on Feature

echo "Login feature code" > login.txt

git add .

git commit -m "Added login feature"

```
D:\git-branching-demo>echo "Login feature code" > login.txt

D:\git-branching-demo>git add .

D:\git-branching-demo>git commit -m "Added login feature"
[feature/login 649007f] Added login feature
1 file changed, 1 insertion(+)
create mode 100644 login.txt

D:\git-branching-demo>
```

Step 6: Merge Feature into Develop git checkout develop

git merge feature/login

```
D:\git-branching-demo>git checkout develop
Switched to branch 'develop'

D:\git-branching-demo>git merge feature/login
Updating cb00c0f..649007f
Fast-forward
 login.txt | 1 +
1 file changed, 1 insertion(+)
create mode 100644 login.txt

D:\git-branching-demo>
```

Step 7: Create Release Branch And Merge into Main

git checkout -b release/v1.0

git checkout main

git merge release/v1.0

```
D:\git-branching-demo>git checkout -b release/v1.0
Switched to a new branch 'release/v1.0'

D:\git-branching-demo>git checkout main
Switched to branch 'main'

D:\git-branching-demo>git merge release/v1.0
Updating cb00c0f..649007f
Fast-forward
 login.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 login.txt
```

Step 9: Hotfix Branch

git checkout -b hotfix/bugfix

Step 10: Merge Hotfix

git checkout main

git merge hotfix/bugfix git checkout develop git merge hotfix/bugfix

```
D:\git-branching-demo>git checkout -b hotfix/bugfix
Switched to a new branch 'hotfix/bugfix'

D:\git-branching-demo>git checkout main
Switched to branch 'main'

D:\git-branching-demo>git merge hotfix/bugfix
Already up to date.

D:\git-branching-demo>git checkout develop
Switched to branch 'develop'

D:\git-branching-demo>git merge hotfix/bugfix
Already up to date.
```

Step 11: Branch Graph

git log --oneline --graph --all

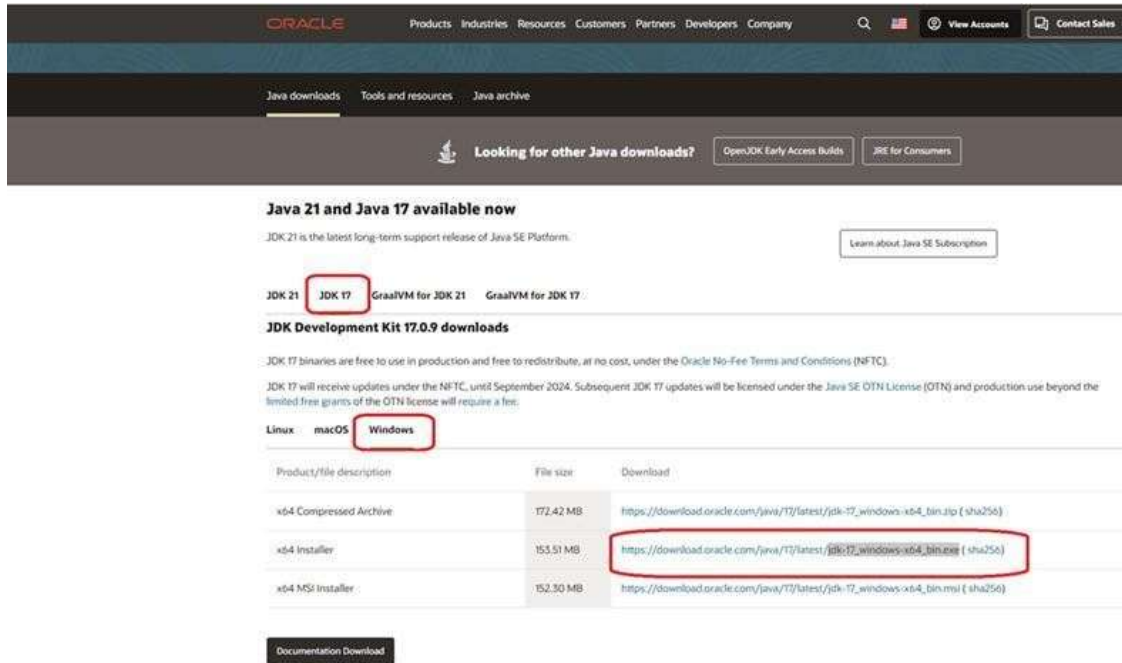
```
D:\git-branching-demo>git log --oneline --graph --all
* 649007f (HEAD -> develop, release/v1.0, main, hotfix/bugfix, feature/login) Added login feature
* cb00c0f Initial commit

D:\git-branching-demo>
```

Practical 2: Use Jenkins to set up a distributed pipeline that will compile and test a Maven project on two different slave nodes respectively.

Step 1: Install Java

Download JDK from: Oracle JDK Downloads



Product/file description	File size	Download
x64 Compressed Archive	172.42 MB	https://download.oracle.com/java/17/latest/jdk-17_windows-x64_bin.zip (sha256)
x64 Installer	153.51 MB	https://download.oracle.com/java/17/latest/jdk-17_windows-x64_bin.exe (sha256)
x64 MSI Installer	152.90 MB	https://download.oracle.com/java/17/latest/jdk-17_windows-x64_bin.msi (sha256)

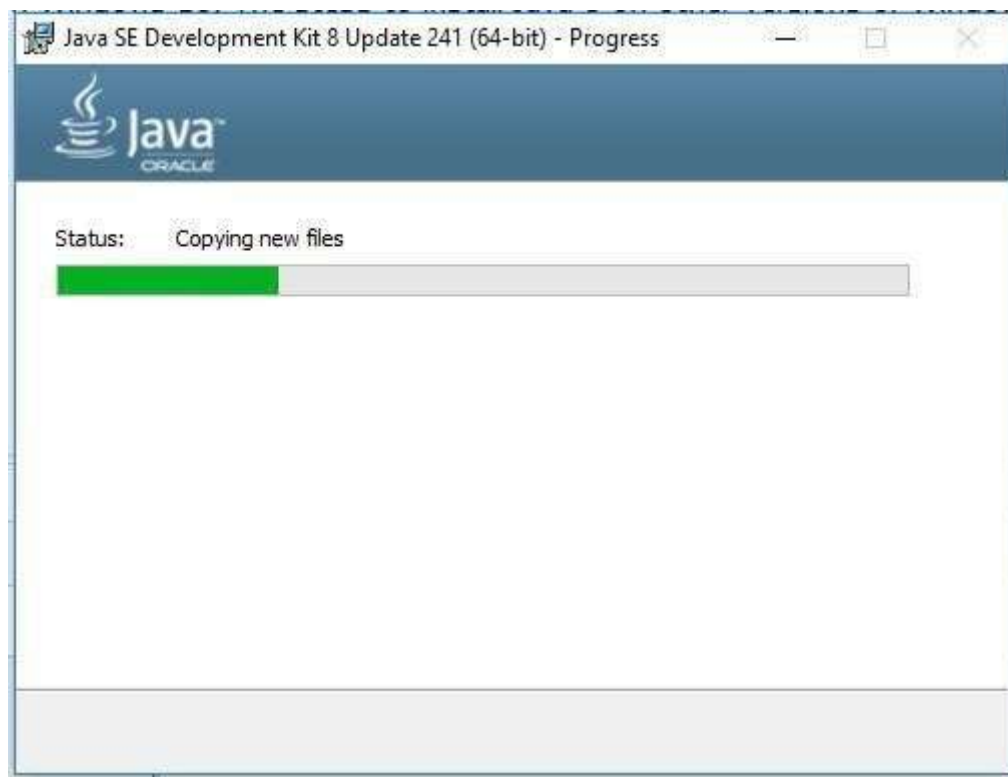
1. **Welcome Screen :** The JDK Setup Wizard opens. Click Next.



- 2. Choose Installation Folder :** Leave the default installation path (recommended) or choose another folder, then click Next.



- 3. Installation in Progress :** Wait while the installer copies the required files.

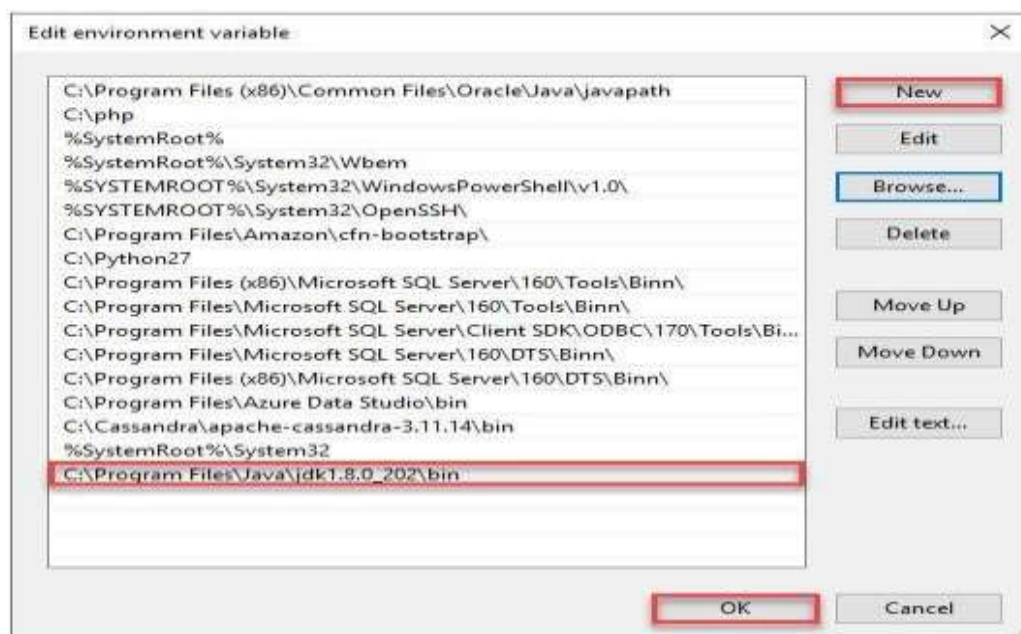


4. Installation Complete: Click Close to finish.



Configure JAVA_HOME

1. Open System Properties → Advanced System Settings.
2. Click Environment Variables.



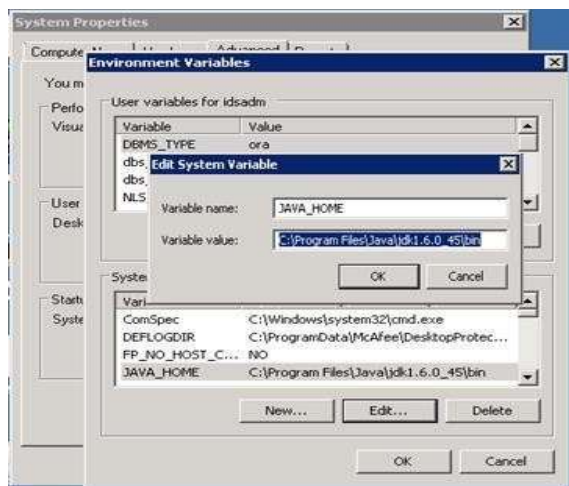
3. Under System Variables, click New.

(Variable Name: JAVA_HOME

Variable Value: C:\Program Files\Java\jdk-21)

4.Edit Path and add:

%JAVA_HOME%\bin



Verify Installation

```

Microsoft Windows [Version 10.0.22631.6199]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>java -version
java version "21.0.2" 2024-01-16 LTS
Java(TM) SE Runtime Environment (build 21.0.2+13-LTS-58)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.2+13-LTS-58, mixed mode, sharing)

C:\Users\Admin>javac -version
javac 21.0.2
  
```

Step 2: Install Apache Maven

Download from: [Apache Maven Official Website](https://maven.apache.org/download.cgi)



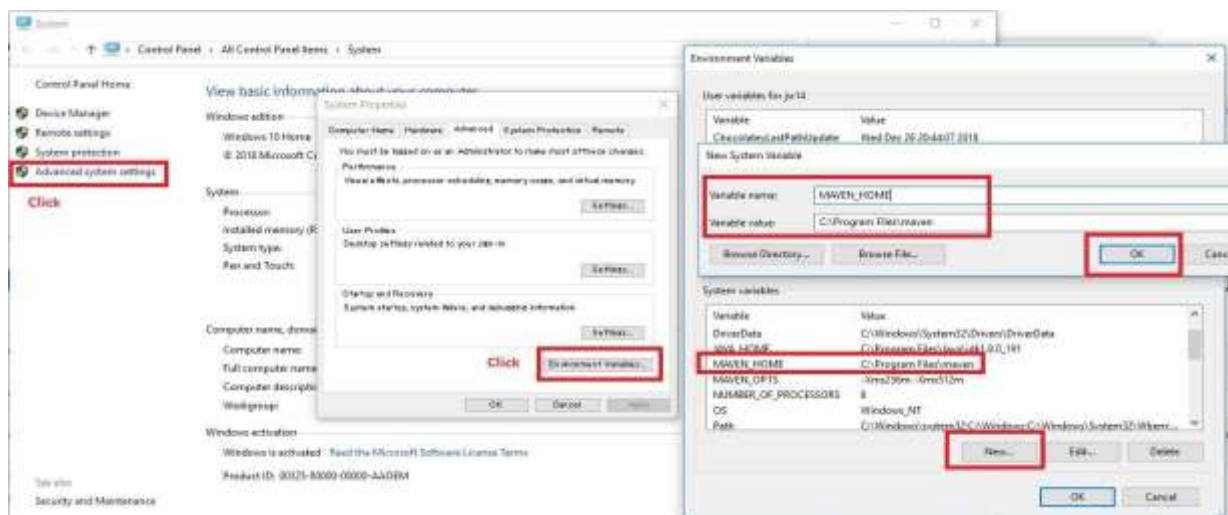
Extract Maven: Extract the ZIP file to:(C:\Program Files\)



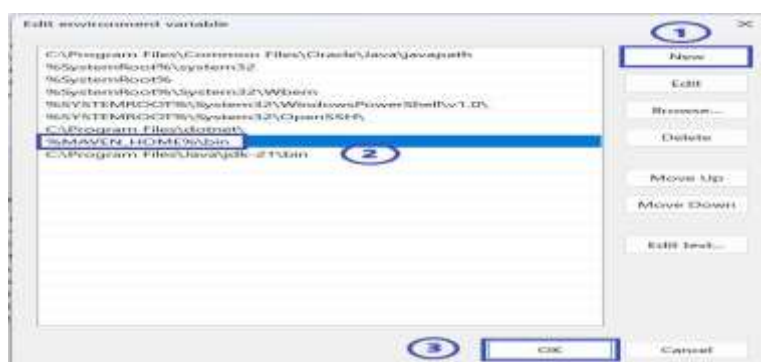
Configure MAVEN_HOME

1. Open Environment Variables.
2. Create a new system variable:

Variable Name: MAVEN_HOME Variable Value: C:\Program Files\Apache\maven



3. Add to Path:
%MAVEN_HOME%\bin

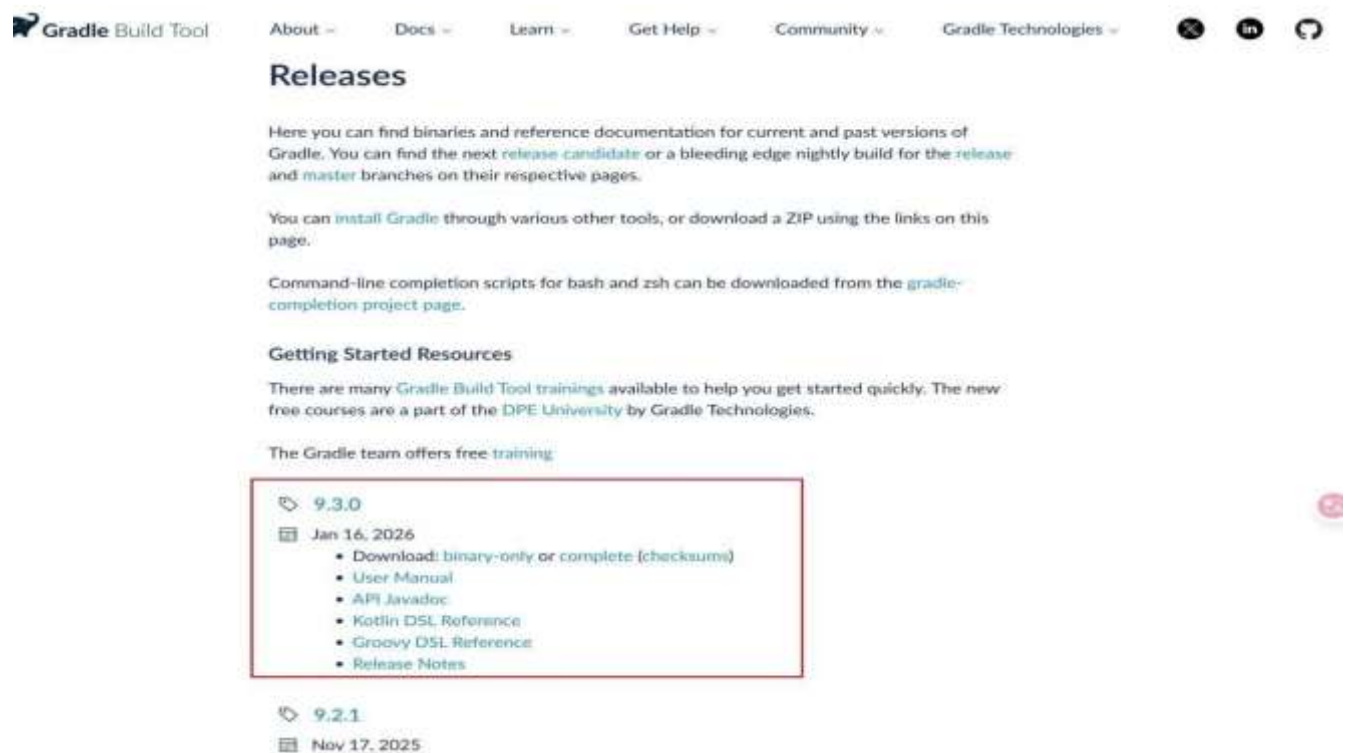


Verify Maven Installation

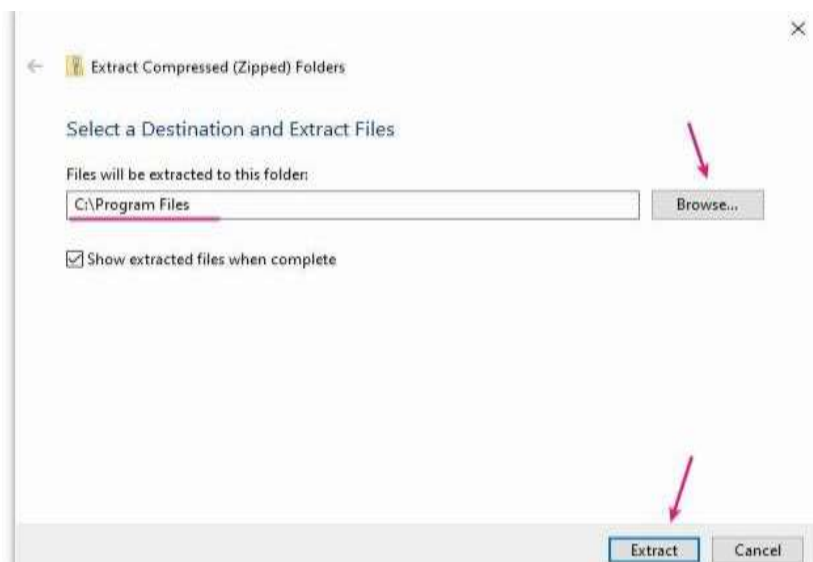
```
C:\Users\Admin>mvn -version
Apache Maven 3.9.16 (2bddd9fddda4b155ebf8000e807eb73fd829a51d5)
Maven home: C:\Program Files\apache-maven-3.9.16
Java version: 21.0.2, vendor: Oracle Corporation, runtime: C:\Program Files\Java\jdk-21
Default locale: en_IN, platform encoding: UTF-8
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"
```

Step 3: Install Gradle

Download Gradle: [Gradle Official Website](https://gradle.org)



Extract Gradle: Extract to (C:\Gradle\gradle-9.x)

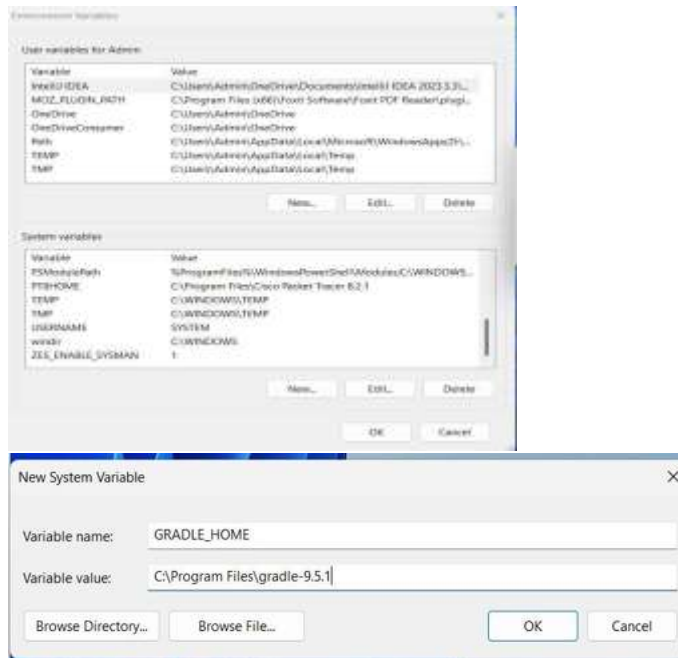


Configure GRADLE_HOME

1. Create a new system variable:(

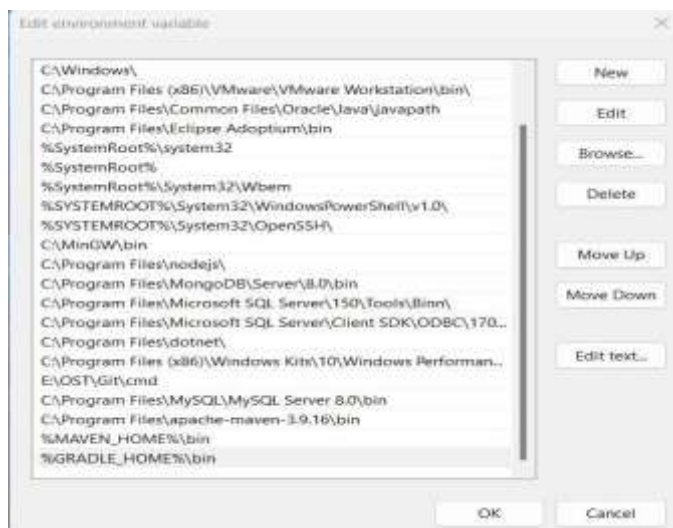
Variable Name: GRADLE_HOME

Variable Value: C:\Gradle\gradle-9.x)



2. Add to Path:

%GRADLE_HOME%\bin



Verify Gradle Installation

```
Microsoft Windows [Version 10.0.22631.6199]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>gradle -v

Welcome to Gradle 9.5.1!

Here are the highlights of this release:
- Task provenance in reports and failure messages
- Type-safe accessors for precompiled Kotlin Settings plugins

For more details see https://docs.gradle.org/9.5.1/release-notes.html

-----
Gradle 9.5.1
-----

Build time: 2026-05-12 13:19:42 UTC
Revision: fd78213f09782e62ca4957f9cf3d90c6c3f1767

Kotlin: 2.3.20
Groovy: 4.0.29
Ant: Apache Ant(TM) version 1.10.15 compiled on August 25 2024
Launcher JVM: 21.0.2 (Oracle Corporation 21.0.2+13-LTS-58)
Daemon JVM: C:\Program Files\Java\jdk-21 (no Daemon JVM specified, using current Java home)
OS: Windows 11 10.0 amd64
```

Step 4: Create a

Maven Project mvn

archetype:generate

-DgroupId=com.example -DartifactId=MavenDemo -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
Microsoft Windows [Version 10.0.22631.6199]
(c) Microsoft Corporation. All rights reserved.

E:\devops\Ex-2>mkdir MavenProjects

E:\devops\Ex-2>cd MavenProjects

E:\devops\Ex-2\MavenProjects>mvn archetype:generate *
More? -DgroupId=com.example *
More? -DartifactId=MavenDemo *
More? -DarchetypeArtifactId=maven-archetype-quickstart *
More? -DinteractiveMode=false
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.1.4/maven-install-plugin-3.1.4.pom (8.1 kB at 5.3 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/43/maven-plugins-43.pom (7.3 kB at 107 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/43/maven-parent-43.pom (58 kB at 553 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/33/apache-33.pom (34 kB at 417 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.10.3/junit-bom-5.10.3.pom (5.6 kB at 118 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.1.4/maven-install-plugin-3.1.4.jar (33 kB at 4.38 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-deploy-plugin/3.1.4/maven-deploy-plugin-3.1.4.pom (9.2 kB at 15.1 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-deploy-plugin/3.1.4/maven-deploy-plugin-3.1.4.jar (48 kB at 545 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-site-plugin/3.12.1/maven-site-plugin-3.12.1.pom (28 kB at 202 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/36/maven-plugins-36.pom (9.9 kB at 148 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml (21 kB at 168 kB/s)
```

Move to project directory: cd MavenDemo

```
E:\devops\Ex-2\MavenProjects>cd MavenDemo

E:\devops\Ex-2\MavenProjects\MavenDemo>dir
Volume in drive E has no label.
Volume Serial Number is 00E0-2984

Directory of E:\devops\Ex-2\MavenProjects\MavenDemo

17-06-2026    08:19    <DIR>          .
17-06-2026    08:19    <DIR>          ..
17-06-2026    08:19                   660 pom.xml
17-06-2026    08:19    <DIR>          src
                   1 File(s)                660 bytes
                   3 Dir(s)      363,777,970,176 bytes free
```

Build Project: mvn clean install

```
C:\Windows\system32\cmd.exe
E:\devops\Ex-2\MavenProjects\MavenDemo>mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] <-- com.example:MavenDemo -->
[INFO] Building MavenDemo 1.0-0-SNAPSHOT
[INFO] from pom.xml
[INFO]
[INFO] [ jar ]
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.4.0/maven-resources-plugin-3.4.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.4.0/maven-resources-plugin-3.4.0.pom (7.9 kB at 5.3 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/45/maven-plugins-45.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/45/maven-plugins-45.pom (6.4 kB at 331 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.4.0/maven-resources-plugin-3.4.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.4.0/maven-resources-plugin-3.4.0.jar (31 kB at 260 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.15.0/maven-compiler-plugin-3.15.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.15.0/maven-compiler-plugin-3.15.0.pom (9.1 kB at 77 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/47/maven-plugins-47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/47/maven-plugins-47.pom (7.2 kB at 98 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/47/maven-parent-47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/47/maven-parent-47.pom (54 kB at 654 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/37/apache-37.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/37/apache-37.pom (38 kB at 333 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.10.2/junit-bom-5.10.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.10.2/junit-bom-5.10.2.pom (3.7 kB at 77 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.15.0/maven-compiler-plugin-3.15.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.15.0/maven-compiler-plugin-3.15.0.jar (84 kB at 884 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-surefire-plugin/3.5.4/maven-surefire-plugin-3.5.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-surefire-plugin/3.5.4/maven-surefire-plugin-3.5.4.pom (9.9 kB at 1.82 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire/3.5.4/surefire-3.5.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire/3.5.4/surefire-3.5.4.pom (19 kB at 242 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/44/maven-parent-44.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/44/maven-parent-44.pom (52 kB at 678 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/34/apache-34.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/34/apache-34.pom (34 kB at 318 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.12.1/junit-bom-5.12.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.12.1/junit-bom-5.12.1.pom (5.6 kB at 77 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-surefire-plugin/3.5.4/maven-surefire-plugin-3.5.4.jar
```

Run: java -cp target\MavenDemo-1.0-SNAPSHOT.jar com.example.App

```
E:\devops\Ex-2\MavenProjects\MavenDemo>java -cp target\MavenDemo-1.0-SNAPSHOT.jar com.example.App
Hello World!
```

Step 5: Create a Gradle Project Create project: gradle init Choose: Application > Java > Groovy DSL

```
C:\Windows\System32\cmd.e x + v
E:\devops\Ex-2>mkdir GradleDemo
E:\devops\Ex-2>cd GradleDemo
E:\devops\Ex-2\GradleDemo>gradle init
Starting a Gradle Daemon (subsequent builds will be faster)

Select type of build to generate:
 1: Application
 2: Library
 3: Gradle plugin
 4: Basic (build structure only)
Enter selection (default: Application) [1..4] 1

Select implementation language:
 1: Java
 2: Kotlin
 3: Groovy
 4: Scala
 5: C++
 6: Swift
Enter selection (default: Java) [1..6] 1

Enter target Java version (min: 7, default: 21): Groovy
Please enter an integer value (min: 7, default: 21):

Project name (default: GradleDemo):

Select application structure:
 1: Single application project
 2: Application and library project
Enter selection (default: Single application project) [1..2]

Select build script DSL:
 1: Kotlin
 2: Groovy
Enter selection (default: Kotlin) [1..2] 2
```

Build project: gradle build

```
E:\devops\Ex-2\GradleDemo>gradle build
Calculating task graph as no cached configuration is available for tasks: build

BUILD SUCCESSFUL in 11s
7 actionable tasks: 6 executed, 1 up-to-date
Configuration cache entry stored.
```

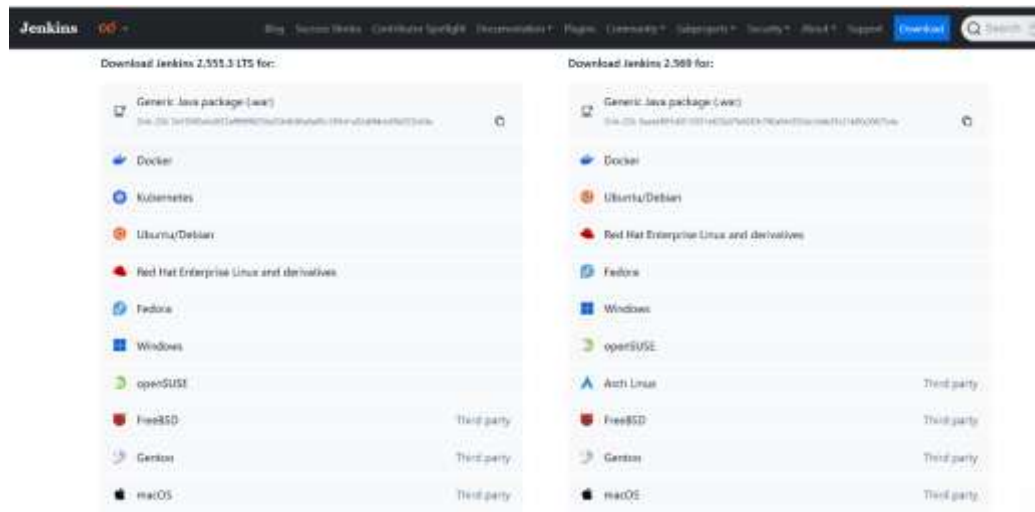
Run application: gradle run

```
E:\devops\Ex-2\GradleDemo>gradle run
Reusing configuration cache.

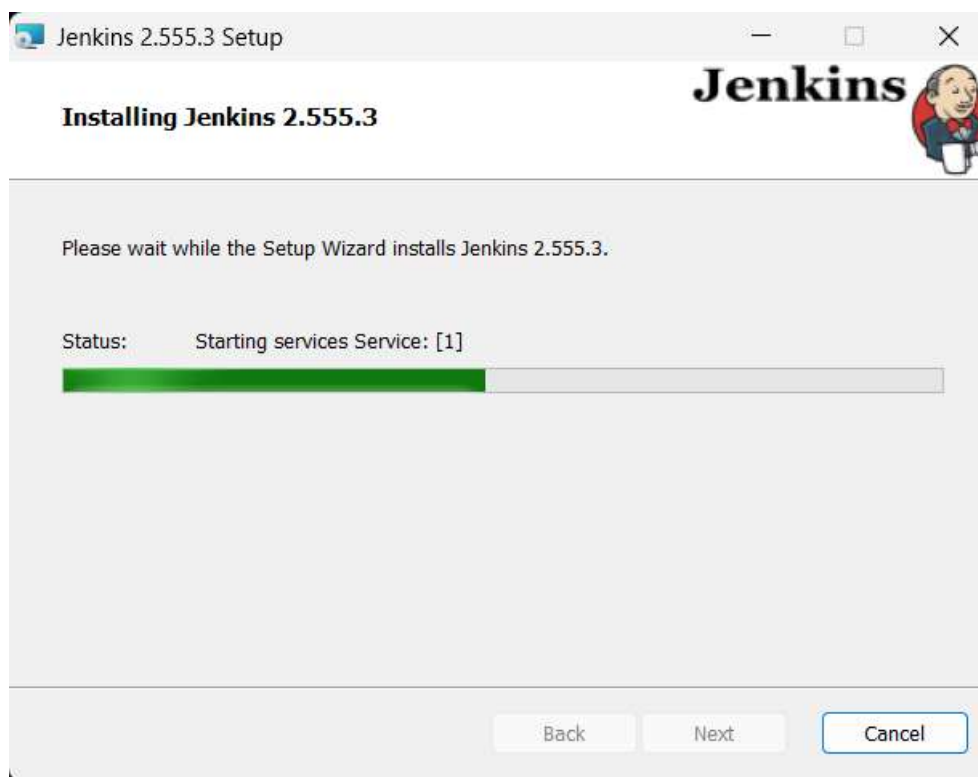
> Task :app:run
Hello World!
```


Practical 3: Create Jenkins job and pipeline it from GitHub repo to run using build.

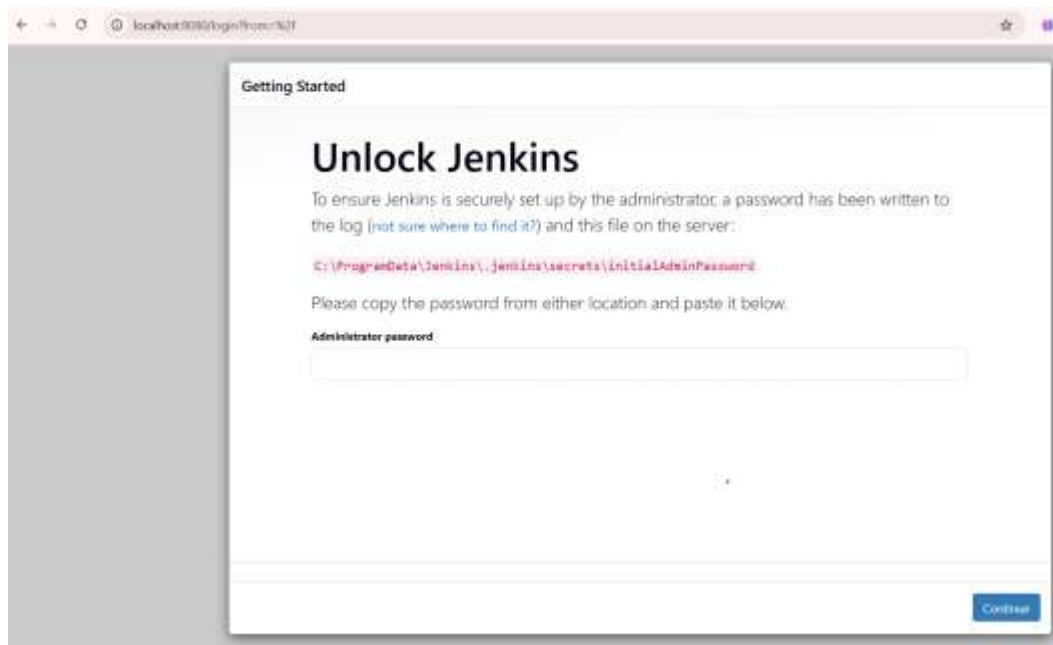
Step 1. Jenkins Install in Your Machine



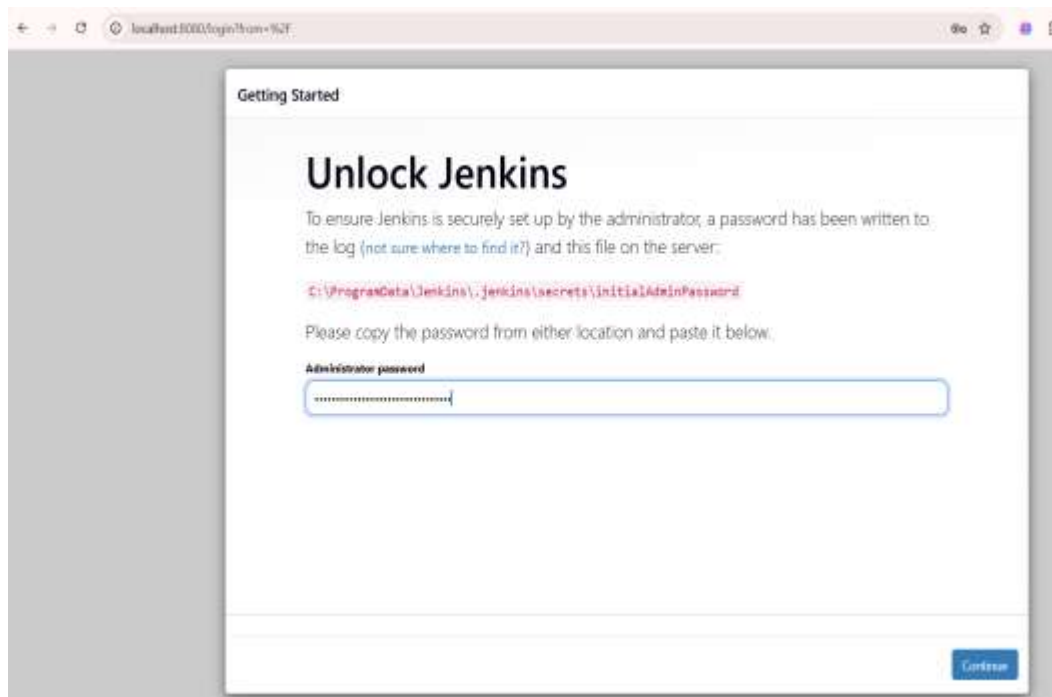
Installed Jenkins:



Step 2. Access Jenkins on Your Browser.- Open your web browser and navigate to the following URL: <http://localhost:8080>



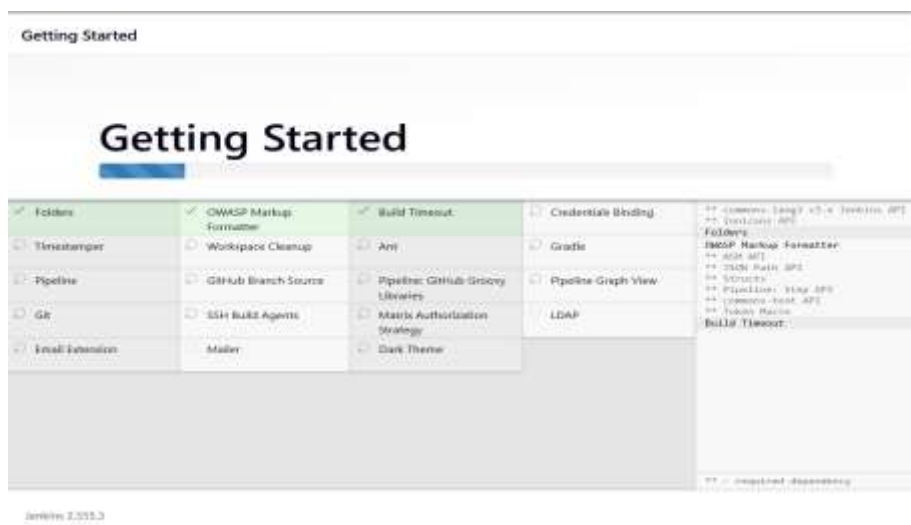
Step 3. Unlock Jenkins Using Generated Administrator Password



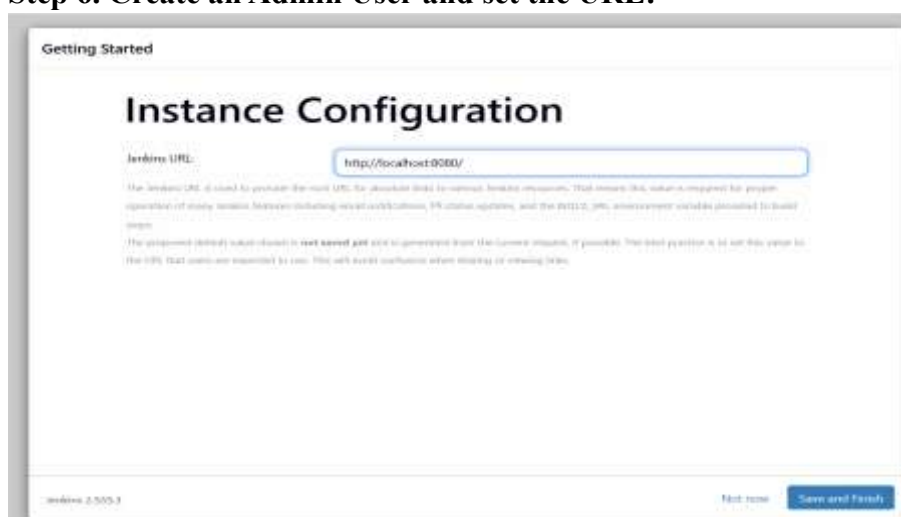
Step 4. Install Suggested Plugins.



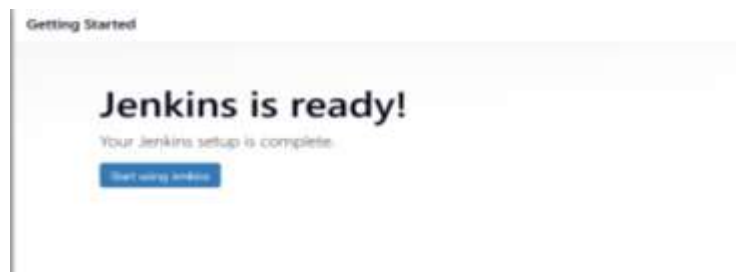
Step 5. Choose Install Suggested Plugins for a quick setup.



Step 6. Create an Admin User and set the URL:



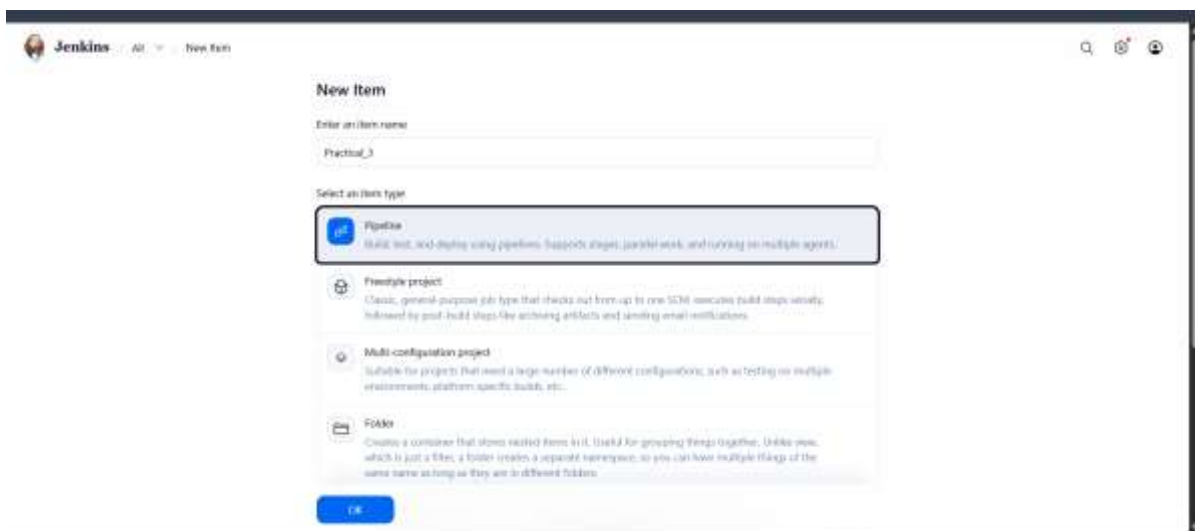
Step 7. Complete Installation



Step 8. Jenkins Localhost dashboard.



Step 9. Create a new Pipeline Using your practical number 3 Select pipeline



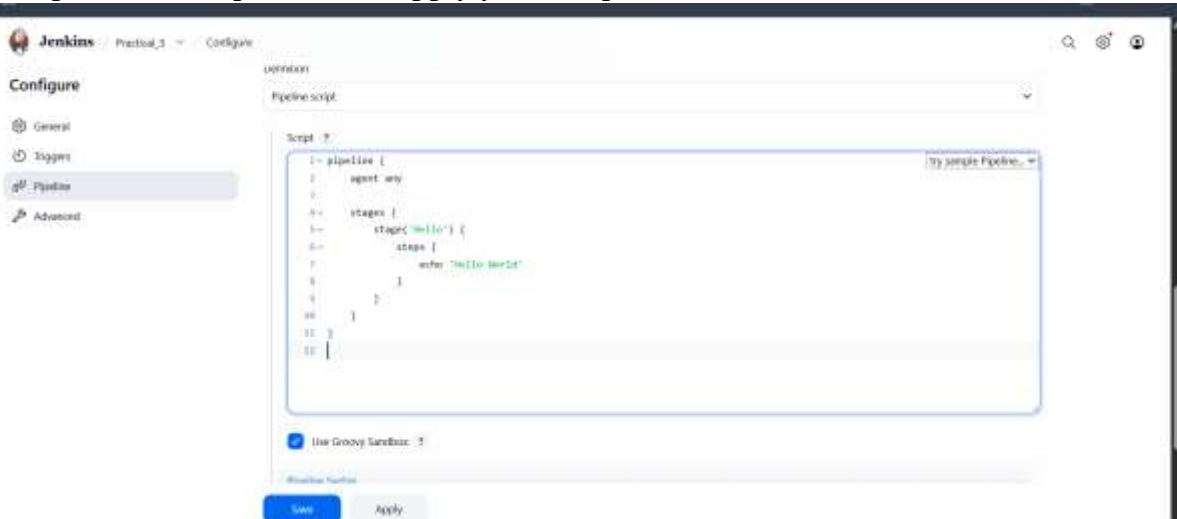
Step 10. Next step Configure your Script (Write Down Basic Script)

```
pipeline {  
    agent any
```

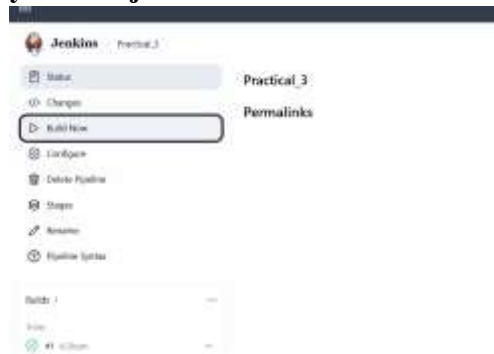
```
    stages {
```

```
stage('Hello') {  
    steps {  
        echo 'Hello World'  
    }  
}  
}
```

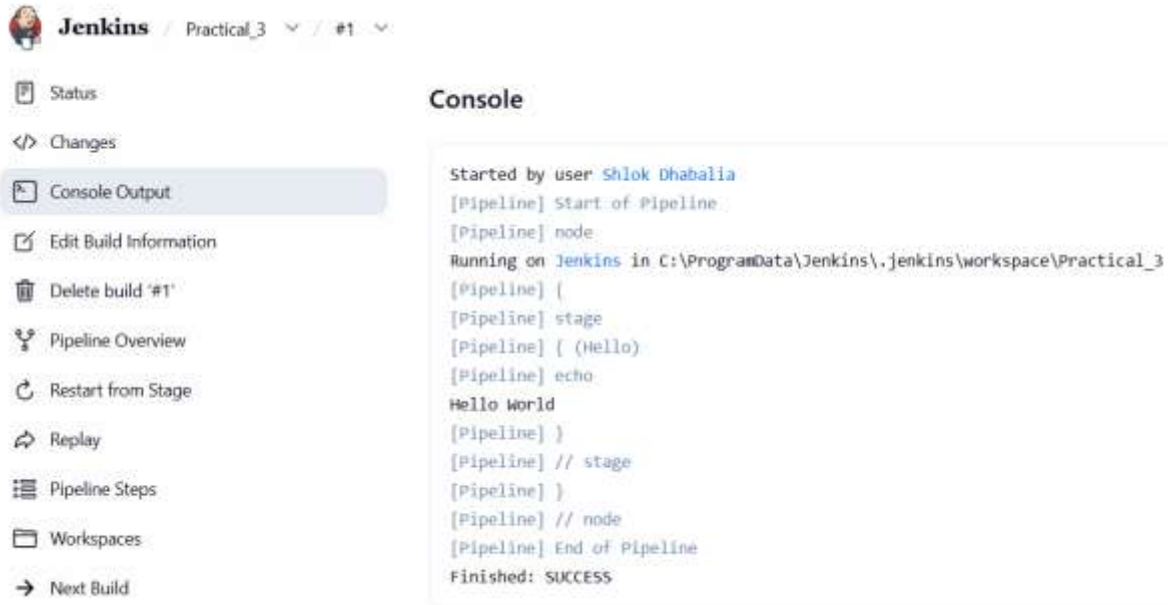
Step 11. Next step Save and Apply your script



Step 12. Next Step Build your Project



Step 13. Next Step Consol Output



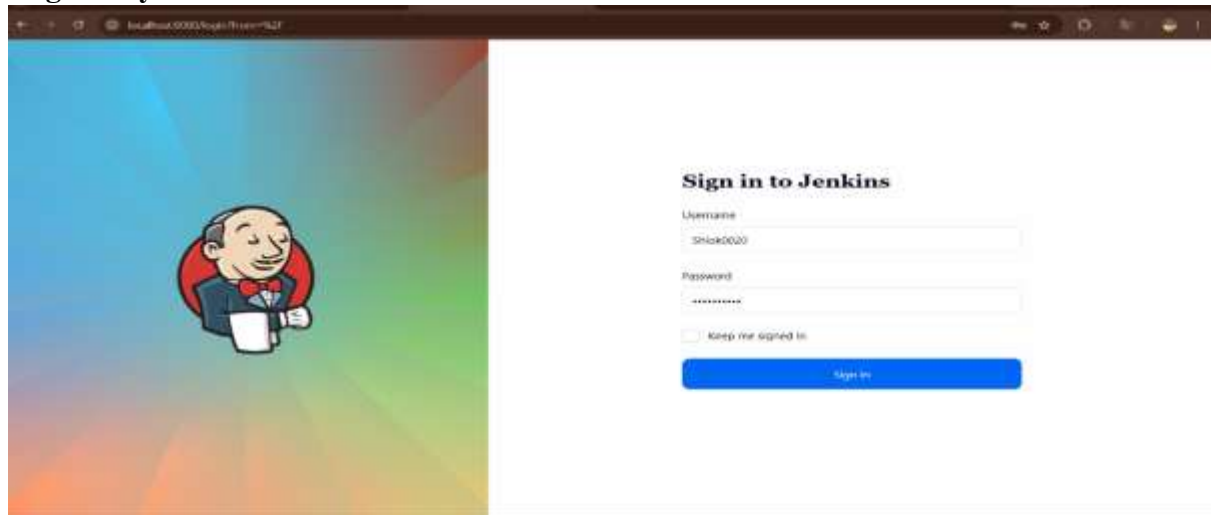
The screenshot shows the Jenkins web interface for a build named 'Practical_3' with ID '#1'. The left sidebar contains navigation links: Status, Changes, Console Output (selected), Edit Build Information, Delete build '#1', Pipeline Overview, Restart from Stage, Replay, Pipeline Steps, Workspaces, and Next Build. The main area displays the console output for the selected build.

Console

```
Started by user Shlok Dhabalia
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\Practical_3
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Hello)
[Pipeline] echo
Hello world
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Practical 4 : Building a CI/CD Pipeline with Jenkins.

Login to your Jenkins localhost:8080/



Install NodeJS plugins.

Dashboard -> Manage-jenkins -> Plugins -> Available plugins and search for NodeJs.



Set Up NodeJs for Project.

Go to Dashboard -> manage Jenkins -> Tools -> NodeJs installation -> add NodeJs and give name to it.



1. Create a new job.

Dashboard > All > New Item

New Item

Enter an item name

Practical - 4

Select an item type

 **Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

 **Pipeline**
Orchestrating long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job types.

OK

2. Now provide github repo link and description.

Jenkins > Practical - 4 > Configure

Configure

General

Description

Repo link: [Theranos](#)

☐ Do not allow builds

☐ Do not allow concurrent builds

☐ Do not allow the pipeline to resume if the controller fails

☒ **GitHub project**
Project url: <https://github.com/Shlok2020/Practical-4.git>

Advanced

Pipeline script visibility: [override](#)

Save Apply

3. Check the github hook trigger.

Dashboard > Practical - 4 > Configuration

Configure

General

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

☐ Build after other projects are built

☐ Build periodically

☒ **GitHub hook trigger for GITScm polling**

☐ Poll SCM

☐ Trigger builds remotely (e.g., from scripts)

Pipeline

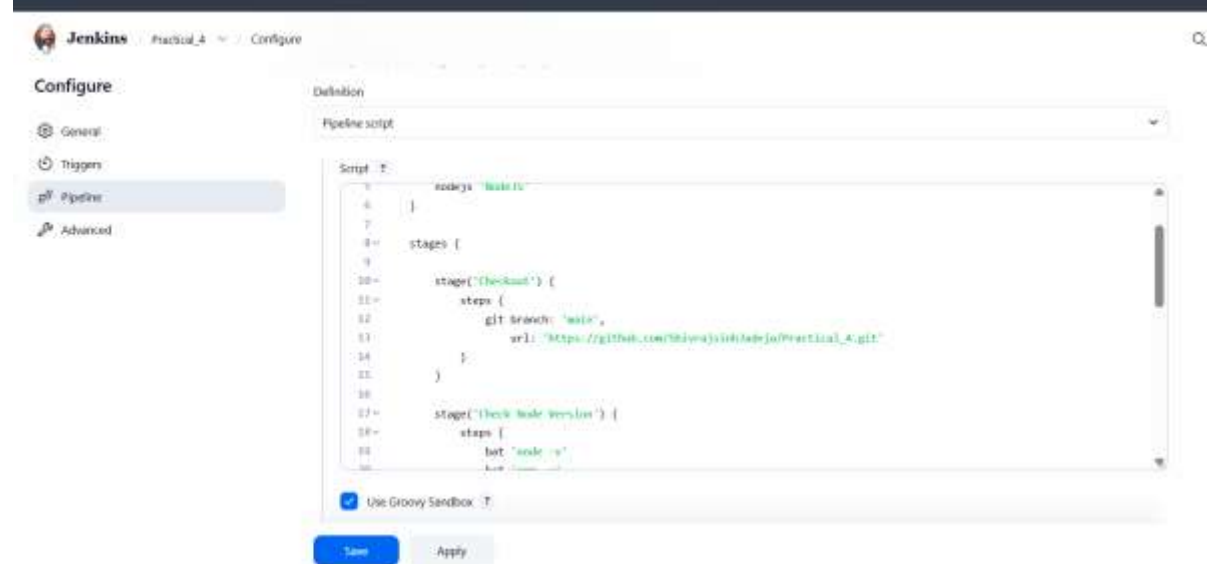
Define your Pipeline using Groovy directly or poll it from source control.

Definition

Pipeline script

Save Apply

Now write the pipeline script.



Here is the script for pipeline:

```

pipeline {
  agent any

  environment {
    NODE_HOME = tool name: 'NodeJS', type: 'nodejs'
    PATH = "${NODE_HOME};${env.PATH}"
  }

  stages {

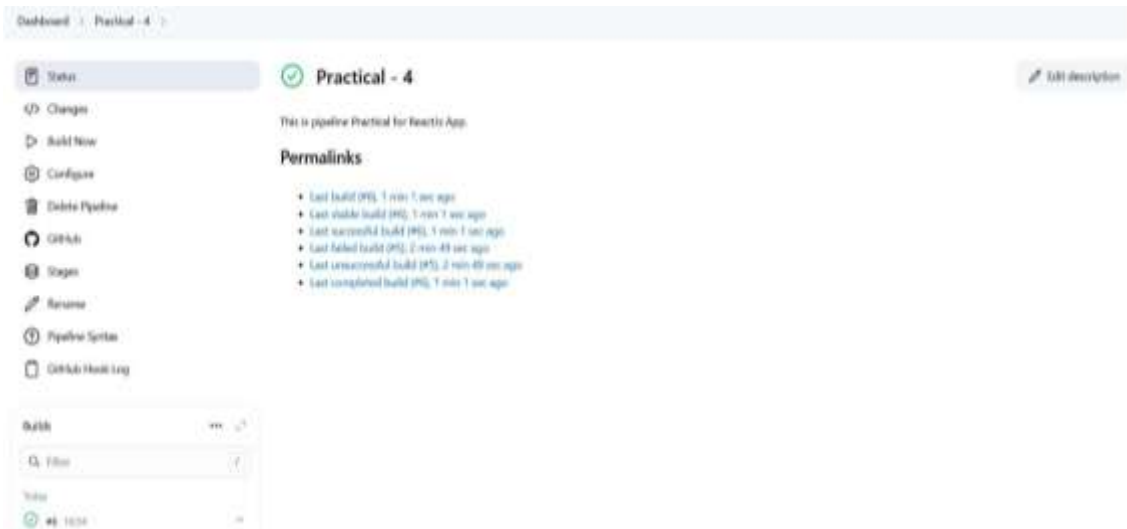
    stage('Checkout') {
      steps {
        git 'https://github.com/Shlok0020/Practical_4.git'
      }
    }

    stage('Check Node Version') {
      steps {
        bat 'node -v'
      }
    }

    stage('Install Dependencies') {
      steps {
        bat 'npm install'
      }
    }
  }
}
  
```

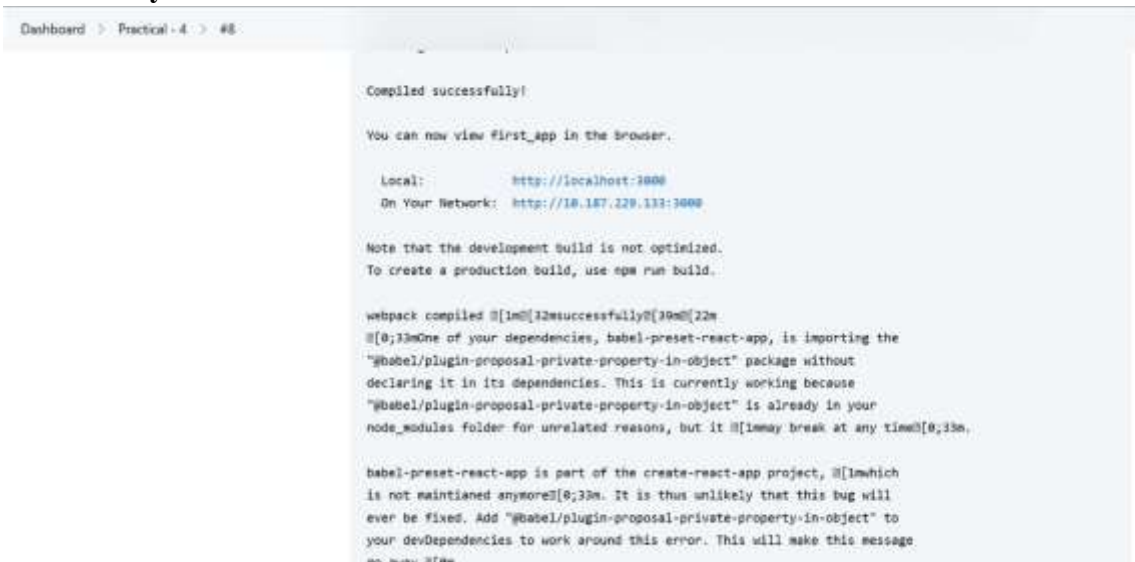
```
stage('Run NodeJS App') {
  steps {
    bat 'npm start'
  }
}
```

Now click on the build .



The screenshot shows the Jenkins interface for a pipeline named 'Practical - 4'. The left sidebar contains navigation links: Status, Changes, Build Now, Configure, Delete Pipeline, GitHub, Stages, Revoke, Pipeline Syntax, and GitHub Hook Log. The main area displays the pipeline status as 'Practical - 4' with a green checkmark. Below this, it says 'This is pipeline Practical for Reactjs App.' and 'Permalink'. A list of build history is shown: Last build (PS), 1 min 1 sec ago; Last stable build (PS), 1 min 1 sec ago; Last successful build (PS), 1 min 1 sec ago; Last failed build (PS), 1 min 49 sec ago; Last unsuccessful build (PS), 2 min 49 sec ago; and Last completed build (PS), 1 min 1 sec ago. At the bottom, there is a 'Build' section with a 'Filter' input and a 'View' button.

Now click on job and goto console output. Now you should see your app running without any issue.



The screenshot shows the Jenkins console output for the 'Practical - 4' pipeline. The breadcrumb navigation at the top reads 'Dashboard > Practical - 4 > #8'. The console output displays the following text:

```
Compiled successfully!

You can now view first_app in the browser.

Local:      http://localhost:3000
On Your Network: http://10.187.229.133:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled @1m@12msuccessfully@39m@22m
@0;13mOne of your dependencies, babel-preset-react-app, is importing the
"@babel/plugin-proposal-private-property-in-object" package without
declaring it in its dependencies. This is currently working because
"@babel/plugin-proposal-private-property-in-object" is already in your
node_modules folder for unrelated reasons, but it @1may break at any time@0;33m.

babel-preset-react-app is part of the create-react-app project, @1which
is not maintained anymore@0;33m. It is thus unlikely that this bug will
ever be fixed. Add "@babel/plugin-proposal-private-property-in-object" to
your devDependencies to work around this error. This will make this message
go away.@0m
```

